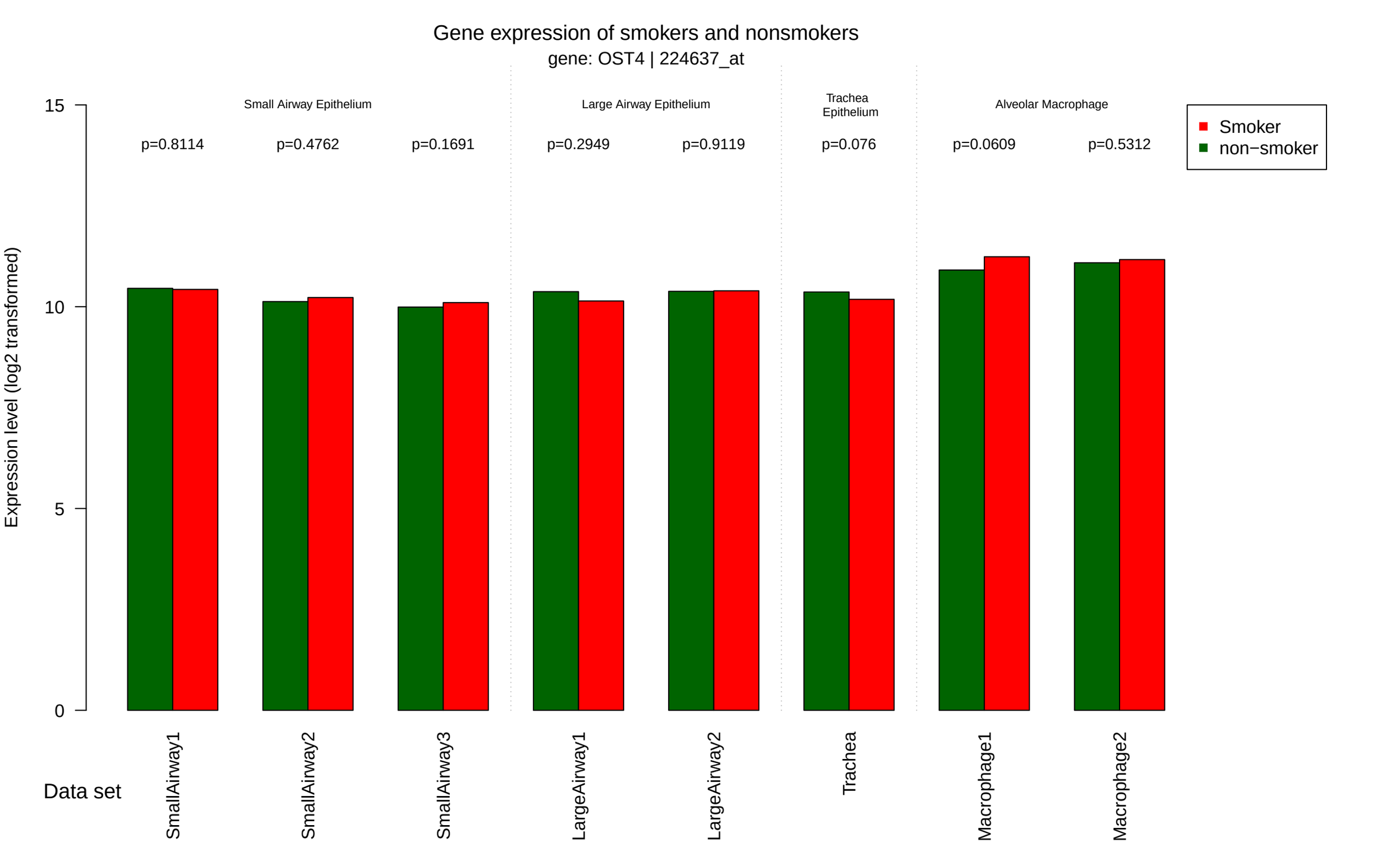
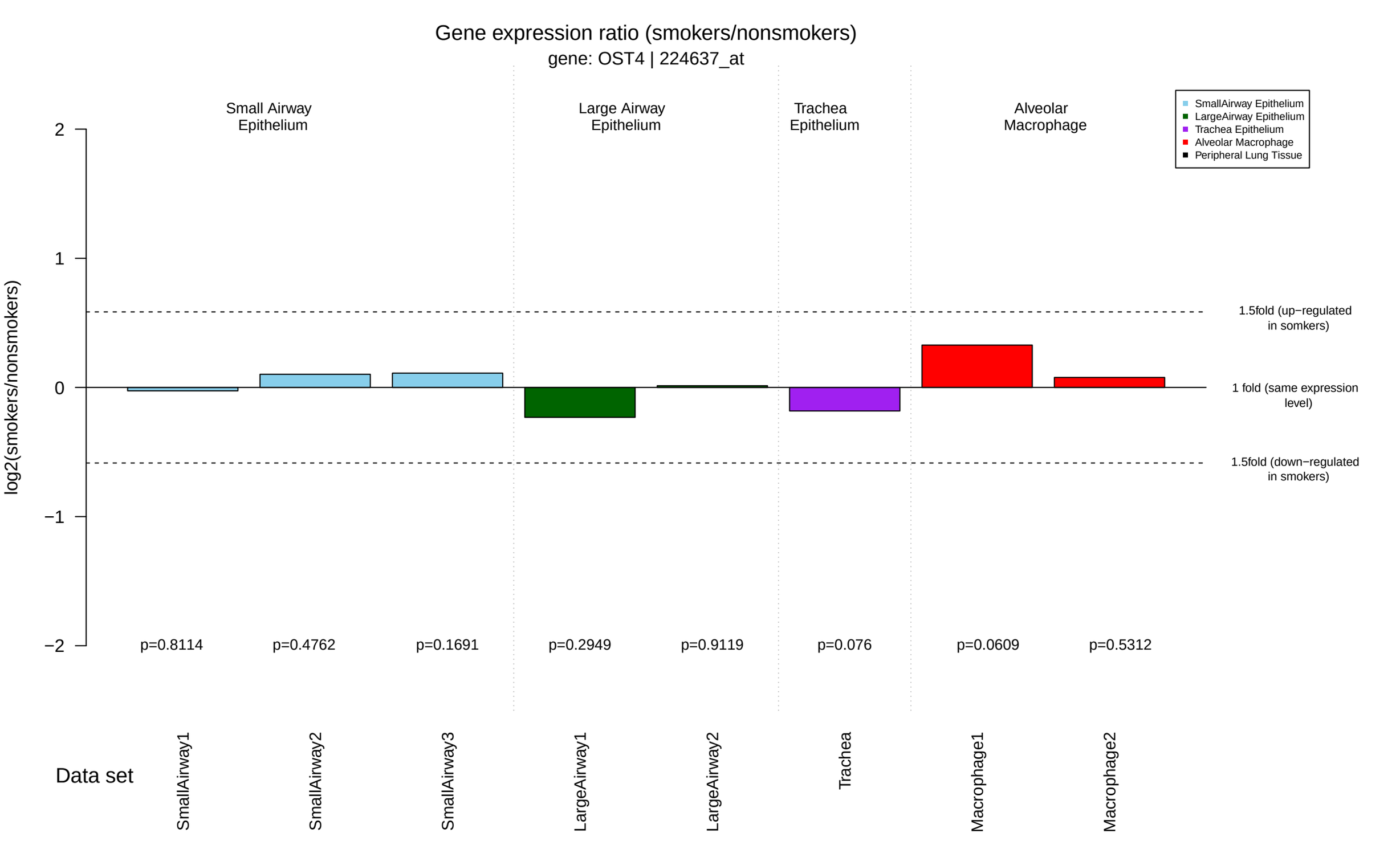
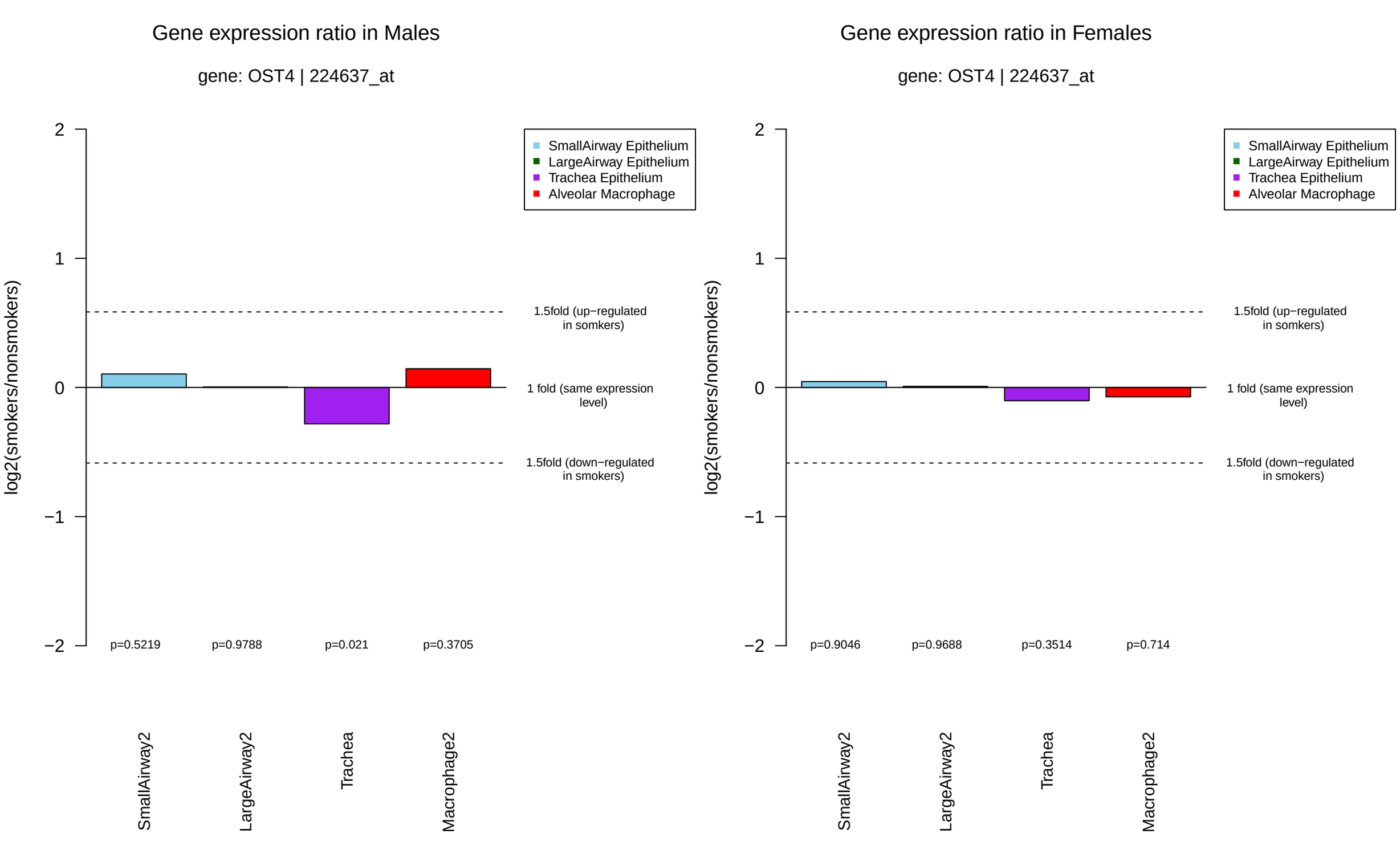
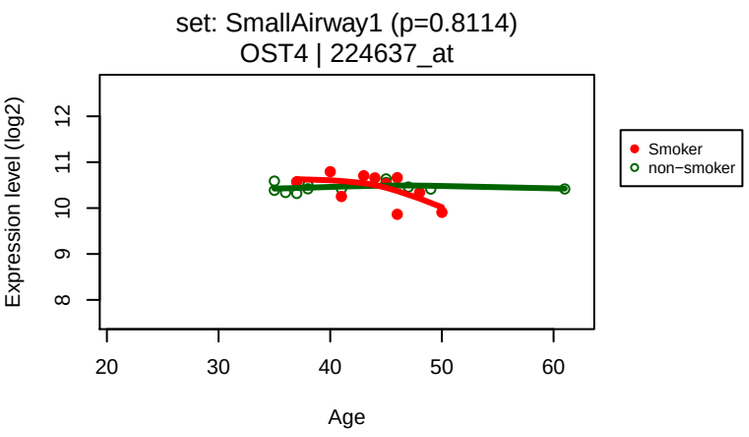
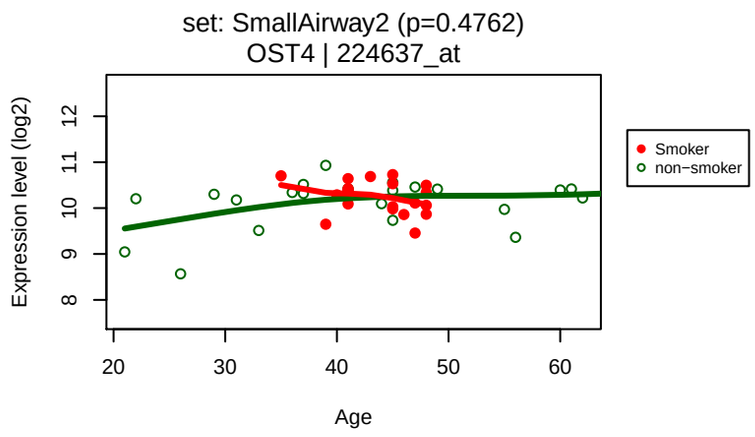
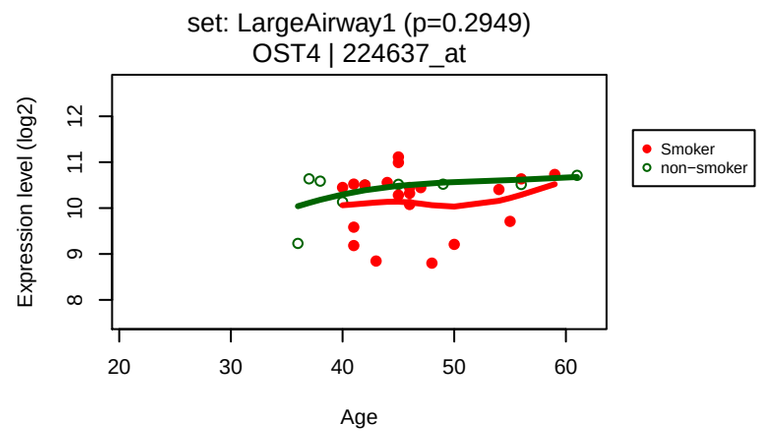

Gene expression of smokers and nonsmokers
gene: OST4 | 224637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8114
p=0.4762
p=0.1691
p=0.2949
p=0.9119
p=0.076
p=0.0609
p=0.5312
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

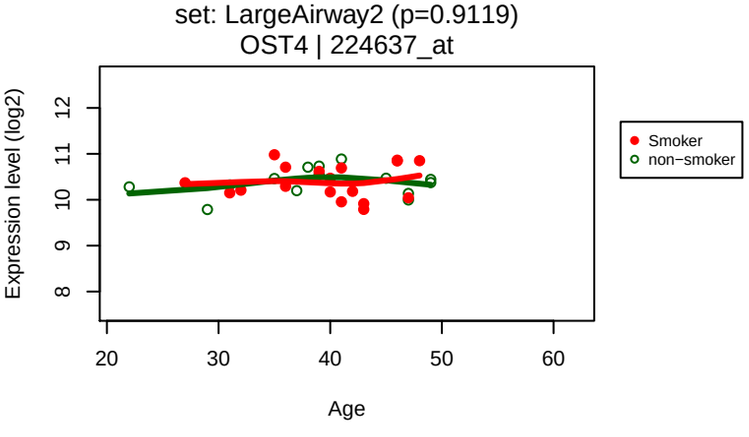

Gene expression ratio (smokers/nonsmokers)
gene: OST4 | 224637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8114
p=0.4762
p=0.1691
p=0.2949
p=0.9119
p=0.076
p=0.0609
p=0.5312
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

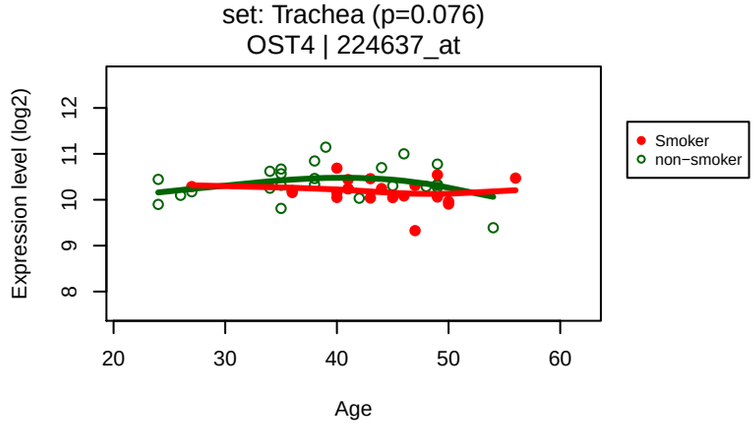

Gene expression ratio in Males
gene: OST4 | 224637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5219
p=0.9788
p=0.021
p=0.3705
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OST4 | 224637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9046
p=0.9688
p=0.3514
p=0.714
SmallAirway2
LargeAirway2
Trachea
Macrophage2

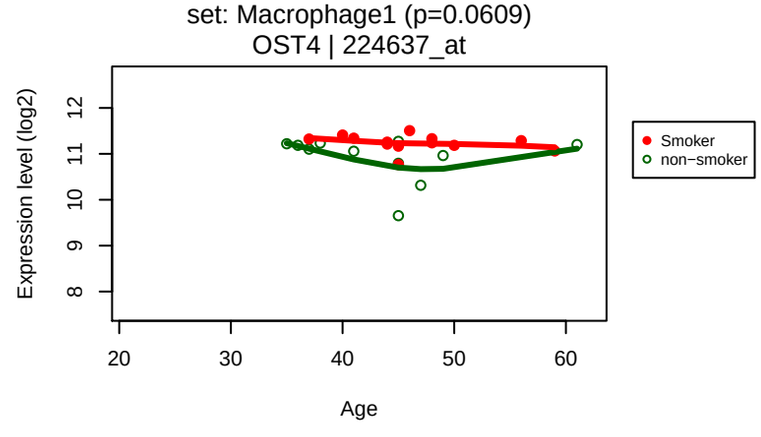

set: SmallAirway1 (p=0.8114)
OST4 | 224637_at
Expression level (log2)
Age
Smoker
non−smoker

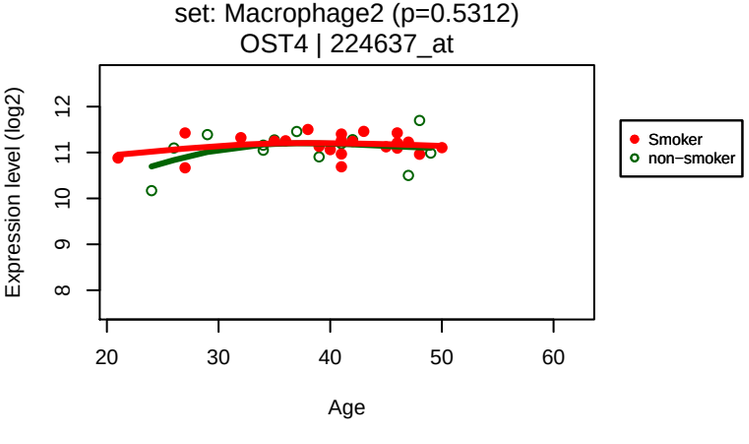

set: SmallAirway2 (p=0.4762)
OST4 | 224637_at
Expression level (log2)
Age
Smoker
non−smoker

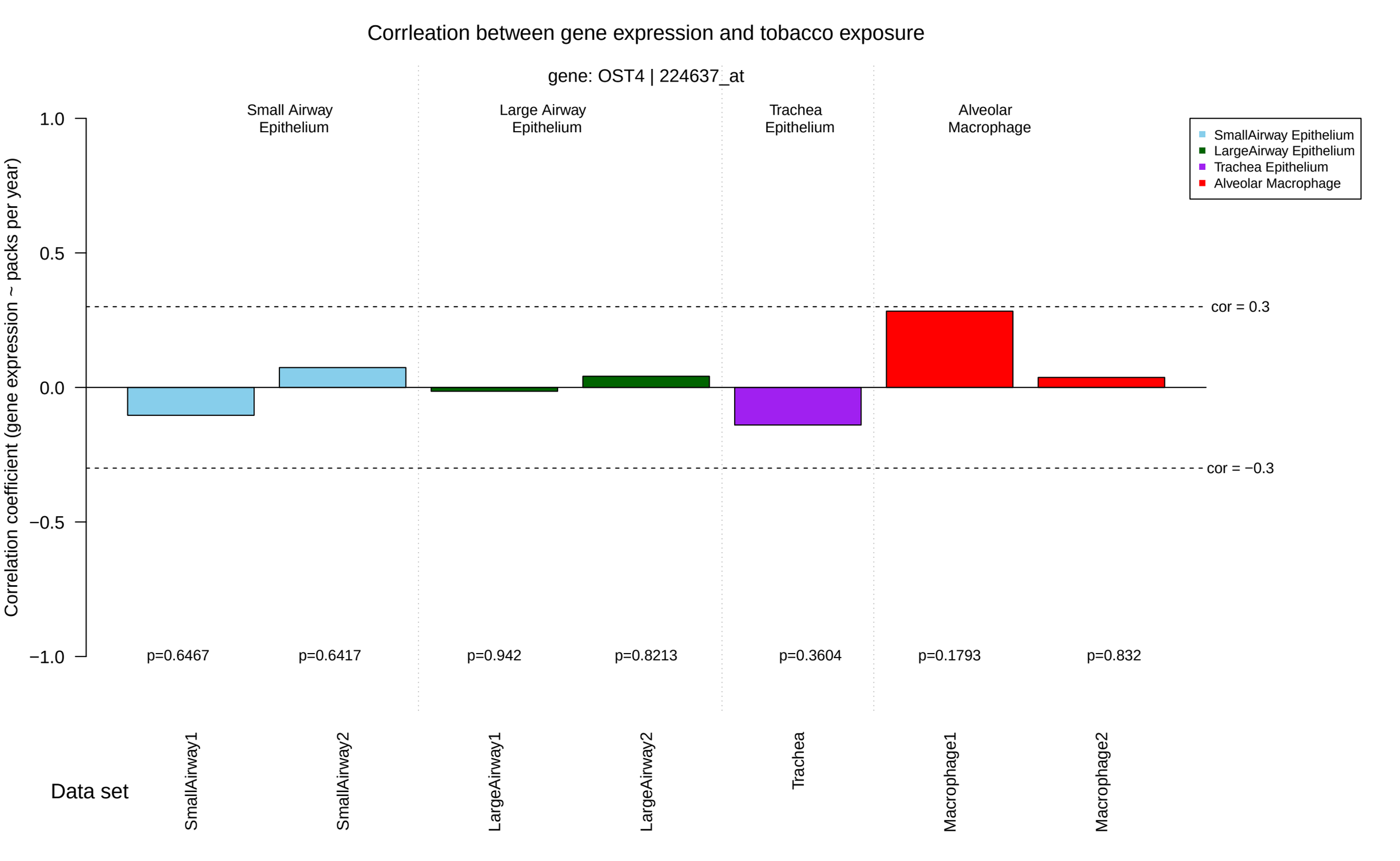

set: LargeAirway1 (p=0.2949)
OST4 | 224637_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9119)
OST4 | 224637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.076)
OST4 | 224637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0609)
OST4 | 224637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5312)
OST4 | 224637_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OST4 | 224637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6467
p=0.6417
p=0.942
p=0.8213
p=0.3604
p=0.1793
p=0.832
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2